

Question	Answer	Marks
5(a)(i)	Reading off graph, at 22.5 °C, $R_T = 1600 \, \Omega$	1

(a)(ii)	Since p.d. across thermistor, $V_{\text{thermistor}} = 2.70 \text{ V} \rightarrow R_T / (R_T + R) \times 9 = 2.70$ $R = 3730 \Omega$	1 1
(b)(i)	Resistance of wire $= \rho L / A = 1.10 \times 10^{-6} \times 1.20 / [\pi \times (0.250 \times 10^{-3})^2]$ $= 6.7227 = 6.72 \Omega$	1 1
(c)(ii)	p.d. across XY, $V_{XY} = [6.7227 / (6.7227 + 1.50 + 6.20)] \times 12.0 = 5.5934 \text{ V}$ At balance point, p.d. across thermistor = p.d. across XJ = 2.70 V Therefore, $XJ = 2.70 / 5.5934 \times 1.2$ $\rightarrow XJ = 0.579 \text{ m}$	1 1
(c)(iii)	When the thermistor is placed at a temperature of 0°C , the resistance of thermistor increases and hence the <u>potential difference across thermistor will increase</u> . As such, <u>the balance point J will be closer to Y</u> .	1 1
(c)(iv)	When the external resistor becomes smaller in value, by potential divider principle, the <u>potential difference across XY becomes larger</u> . As such, the <u>balance point J will be closer to X</u> .	1 1
	Max Marks	11

Question	Answer	Marks
24. (ii)	Balance point is defined as the point of zero current to null deflection.	1